Project Upgrade Report

Frontend, grounded medical AI, dataset analytics, LoRA preparation, deployment hardening, and end-to-end quality assurance.

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Repository: Ganesh172919/React-App-LLM

SOURCE ROWS	CITATION COVERAGE	TRAINING RUNS
16,412	100%	0
Dataset profiled	Live fallback QA	Preparation only

OUTCOME

A deployment-ready full-stack medical assistant foundation

- React chat workspace with persistent CRUD, export, search, source cards, and analytics.
- Node API with authenticated WebSockets, safety screening, PubMed grounding, citation checks, and health reporting.
- Reproducible dataset analysis and LoRA data preparation with training explicitly disabled.

EXECUTIVE SUMMARY

What changed and why it matters

The project was upgraded from a basic chat prototype into an inspectable product foundation. The main change is separation of concerns: browser-safe UI behavior, server-side AI orchestration, source-grounded medical retrieval, persistent user-owned history, and reproducible data tooling now have explicit boundaries.

DELIVERED SCOPE

- Responsive landing page, authenticated chat workspace, and analytics dashboard.
- Create, read, rename, search, export, and delete flows for user-owned conversations.
- Medical routing through PubMed with visible references and algorithmic citation validation.
- History-aware AI context, harmful-message filtering, rate limits, and private WebSocket updates.
- Dataset profile, duplicate audit, deterministic preparation, split verification, and tests.
- No-training LoRA config validation plus SFT JSONL preparation utilities.
- Same-origin container topology with Nginx API/WebSocket proxying and service health checks.
- Expanded documentation, CI workflow, author attribution, and release report.

MEASURED RESULTS

BACKEND TESTS	PYTHON TESTS	NPM FINDINGS	FRONTEND JS
11	6	0	397 KB
All passing	Dataset + LoRA	Production audit	124.6 KB gzip

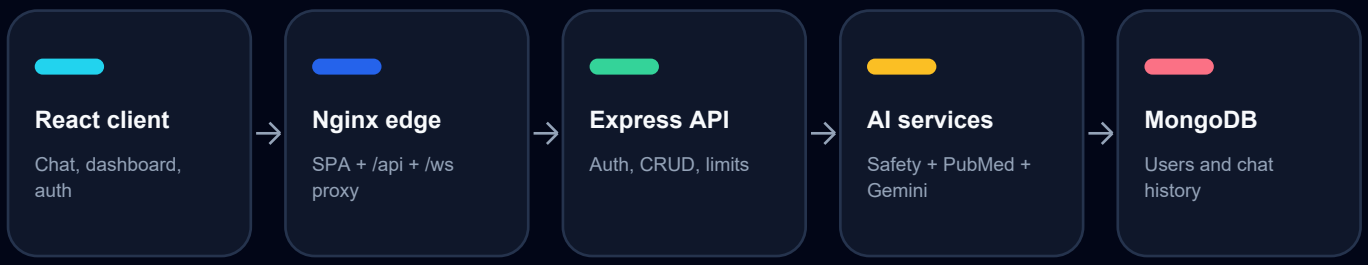
IMPORTANT BOUNDARY

The data reports are structural evidence, not clinical validation. The LoRA package validates configuration and prepares records only; no model was downloaded, trained, benchmarked, or published during this upgrade.

SYSTEM DESIGN

End-to-end architecture

One public frontend origin keeps cookies, API calls, and WebSockets consistent in production. AI provider credentials remain server-side.



MEDICAL REQUEST PATH

01 Screen
Validate size and detect unsafe or urgent content.

02 Normalize
Reduce conversational wording to focused PubMed terms.

03 Retrieve
Search and summarize up to three relevant PubMed records.

04 Generate
Use optional Gemini generation bounded to retrieved records.

05 Verify
Map [n] citations to supplied sources and compute coverage.

06 Persist
Save response, sources, safety state, and verification metadata.

TRUST BOUNDARIES

- HTTP-only cookies protect browser sessions; authenticated user IDs scope history and WebSocket broadcasts.
- API keys are never placed in Vite variables or browser bundles; provider failure degrades to source-only medical responses.

PRODUCT AND UX

Conversation workspace and dashboard

PRIMARY USER JOURNEY

01	**Register / sign in**	Cookie session is restored before route decisions, preventing authenticated users from flashing or becoming trapped on login.
02	**Ask a question**	Textarea supports clear focus states, Enter-to-send, Shift+Enter newline, character count, and disabled loading state.
03	**Review evidence**	Medical replies show source cards, verification status, citation coverage, and a professional-care reminder.
04	**Manage history**	Search, rename, export, reload, and delete controls are grouped per saved conversation.
05	**Inspect analytics**	Selectable 7, 30, or 90-day activity plus verification and dataset quality are visible and exportable.

LIVE BROWSER QA

Medical message	**Pass - PubMed references visible**
Citation coverage	**Pass - 100% for source-only fallback**
History controls	**Pass - search, rename, export, reload**
Dashboard	**Pass - metrics, period filters, export, refresh**
Responsive layout	**Pass - 390 px width, no horizontal overflow**
Session lifecycle	**Pass - restore, WebSocket reconnect, logout**

DATASET ANALYSIS

Medical Q&A descriptive profile

SOURCE ROWS	VALID PAIRS	UNIQUE QUESTIONS
16,412	16,407	14,345
CSV input	5 missing answers	Normalized text

ANSWER LENGTH DISTRIBUTION



QUESTION CATEGORIES



INTERPRETATION

Median answer length is 132 words and the 95th percentile is 489. Definition-style prompts dominate the dataset, so evaluation should be stratified by question type rather than relying on one aggregate score.

DATASET AUDIT

Conflicts, provenance, and review risk

MULTI-ANSWER GROUPS	LOW SIMILARITY	MEAN SIMILARITY
1,396	1,128	0.192
Distinct answers	Below 0.35	Token Jaccard

STRUCTURAL FLAGS

Missing answer rows	5
Exact normalized duplicates	48
Repeated question rows	2,062
Answers containing URLs	159
Answers with non-ASCII characters	61

EXAMPLES REQUIRING REVIEW

What is (are) High Blood Cholesterol ?

19 answers | mean similarity 0.133 | answer range 20-752 words

What is (are) Medicare and Continuing Care ?

14 answers | mean similarity 0.176 | answer range 70-615 words

What are the treatments for Breast Cancer ?

13 answers | mean similarity 0.139 | answer range 60-4,133 words

RECOMMENDATION

Keep answer variants and their provenance visible. Do not silently merge conflicting clinical claims. Complete licensing and qualified clinical review before retrieval benchmarking or future model training.

LORA PREPARATION

Validated configuration - no training

The retired Colab export was replaced with a safe package that validates adapter settings, estimates adapter size, and creates deterministic SFT JSONL records. It cannot train or publish a model in its committed configuration.

CONFIGURATION

Base model	**gpt2**
Task type	**CAUSAL_LM**
Rank / alpha	**8 / 32**
Dropout	**0.05**
Target modules	**c_attn**
Maximum sequence length	**512**
Estimated adapter parameters	**147,456**
Estimated FP16 size	**0.281 MB**

PREPARATION WORKFLOW

1 Analyze

Profile rows, lengths, categories, and missing values.

2 Audit

Review duplicate questions and low-similarity answer variants.

3 Split

Create deterministic question-level train, validation, and test files.

4 Convert

Write prompt-answer JSONL plus hashes and record counts.

SAFETY GATE

- training_enabled is false and publish_adapter is false.
- Future training requires dataset licensing, clinician review, held-out evaluation, privacy review, and an approved release process.

BACKEND AND AI

Safety, grounding, and persistence

Prompt safety

Input size limits, self-harm redirection, possible-emergency notices, and harmful-instruction screening.

Medical retrieval

Focused PubMed query normalization, ESearch and ESummary requests, visible PMIDs, titles, journals, and links.

Citation audit

Valid and invalid [n] markers, cited-source coverage, grounding state, provider, and model metadata.

History context

Authenticated recent-chat retrieval with lightweight token-overlap scoring and explicit metadata on reused messages.

Private updates

WebSocket clients are authenticated from HTTP-only cookies and broadcasts are filtered to the owning user.

Operations

Request IDs, security headers, JSON limits, AI rate limits, structured 404/errors, graceful shutdown, and health status.

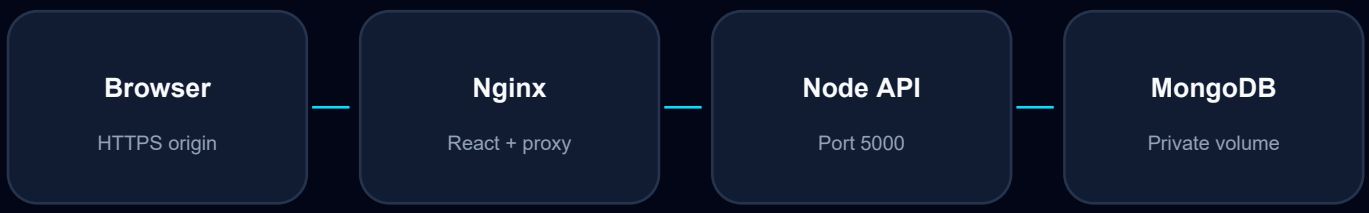
FAILURE BEHAVIOR

Gemini key missing	**Return PubMed references only; do not invent an answer**
No PubMed match	**Return explicit fallback signal to the client**
Unsafe prompt	**Return safety redirect metadata without generation**
Database disconnected	**Expose state in /api/health; database routes fail clearly**
WebSocket interruption	**Exponential reconnect without losing HTTP functionality**

DEPLOYMENT

Same-origin container topology

Public request



PRODUCTION CONFIGURATION

PUBLIC_ORIGIN	**Exact HTTPS frontend origin**
COOKIE_SECURE	**true behind HTTPS**
JWT_* secrets	**Four unique high-entropy values**
MONGO_URI	**Private MongoDB connection string**
GEMINI_API_KEY	**Optional server-side generation key**
BREVO_API_KEY	**Optional email delivery key**

RELEASE CHECKLIST

- Copy .env.docker.example to .env and replace every placeholder secret.
- Set PUBLIC_ORIGIN and COOKIE_SECURE=true for the final HTTPS hostname.
- Run docker compose up --build -d and wait for MongoDB, API, and frontend health checks.
- Verify /api/health, login cookies, WebSocket connection, medical source links, and dashboard metrics.
- Configure backups, log retention, monitoring, and an incident response owner before public release.

Local command

`docker compose up --build -d`

Docker CLI was unavailable in this QA environment; configuration is included for target-host validation.

VALIDATION AND ROADMAP

Release evidence and next controls

AUTOMATED EVIDENCE

Backend unit tests	**11 passing - citation, history, PubMed query, safety**
Python unit tests	**6 passing - dataset and LoRA tools**
Frontend lint	**Passing - ESLint**
Frontend production build	**Passing - 217 modules, 124.6 KB JS gzip**
Dependency audit	**0 production vulnerabilities in frontend and backend**
Python compilation	**Passing - all maintained dataset and LoRA modules**

KNOWN LIMITATIONS

- Citation-number validation proves mapping to retrieved records, not the clinical truth of every generated statement.
- The custom safety filter is a first layer; production needs provider safety controls, abuse monitoring, and escalation procedures.
- The dataset lacks complete per-answer provenance and licensing fields, and multi-answer questions need qualified review.
- Conversation retrieval is lexical token overlap, not a production semantic vector index.

RECOMMENDED NEXT STEPS

- Deploy to a staging HTTPS origin and execute the browser QA checklist against production containers.
- Add mocked controller integration tests plus a repeatable browser automation suite for auth, CRUD, chat, and analytics.
- Introduce structured observability, secret rotation, database backups, and uptime alerting.
- Complete clinical, privacy, accessibility, and dataset licensing reviews before any healthcare-facing release.

PROJECT ATTRIBUTION

J Ganesh kumar reddy

Full-stack product work spanning React UI/UX, backend CRUD and authentication, medical AI integration, dataset curation, safeguards, analytics, testing, documentation, and deployment preparation.